Developer manual

> 💡 The RayNeo developer manual can help you quickly understand the basic situation and ability boundaries of the latest X-series glasses, and help you solve various problems encountered in the development process
>
> This document is mainly targeted at the RayNeo X series glasses (X2, X3 Pro).
>
> 👓**Quickly get to know RayNeoX2**
>
> TCL Rayneo X2 AR Glasses Review
>
> 🕵️**How to Install Apps**
>
> web installation tool

Relevant materials

RayNeo Official Website

Open Platform

Development Documents

- 📄Device Introduction
- 📄Design specifications for AR glasses
- 📄OpenXR unity ARDK
- 📄ARDK for Android
- 📄Debugging Tools
- 📄FAQ

Discord： **join us**

Important Update Notes

1. **September 26, 2025**

[Unity SDK] For versions of Glass OS above 23.8.29, please use Unity SDK 1.1.2 for development ARDK Download

2. **August 13, 2025**

[ADB Switch] For versions of X3pro Glass OS above 25.8.13, an ADB switch has been added. Please check ADB

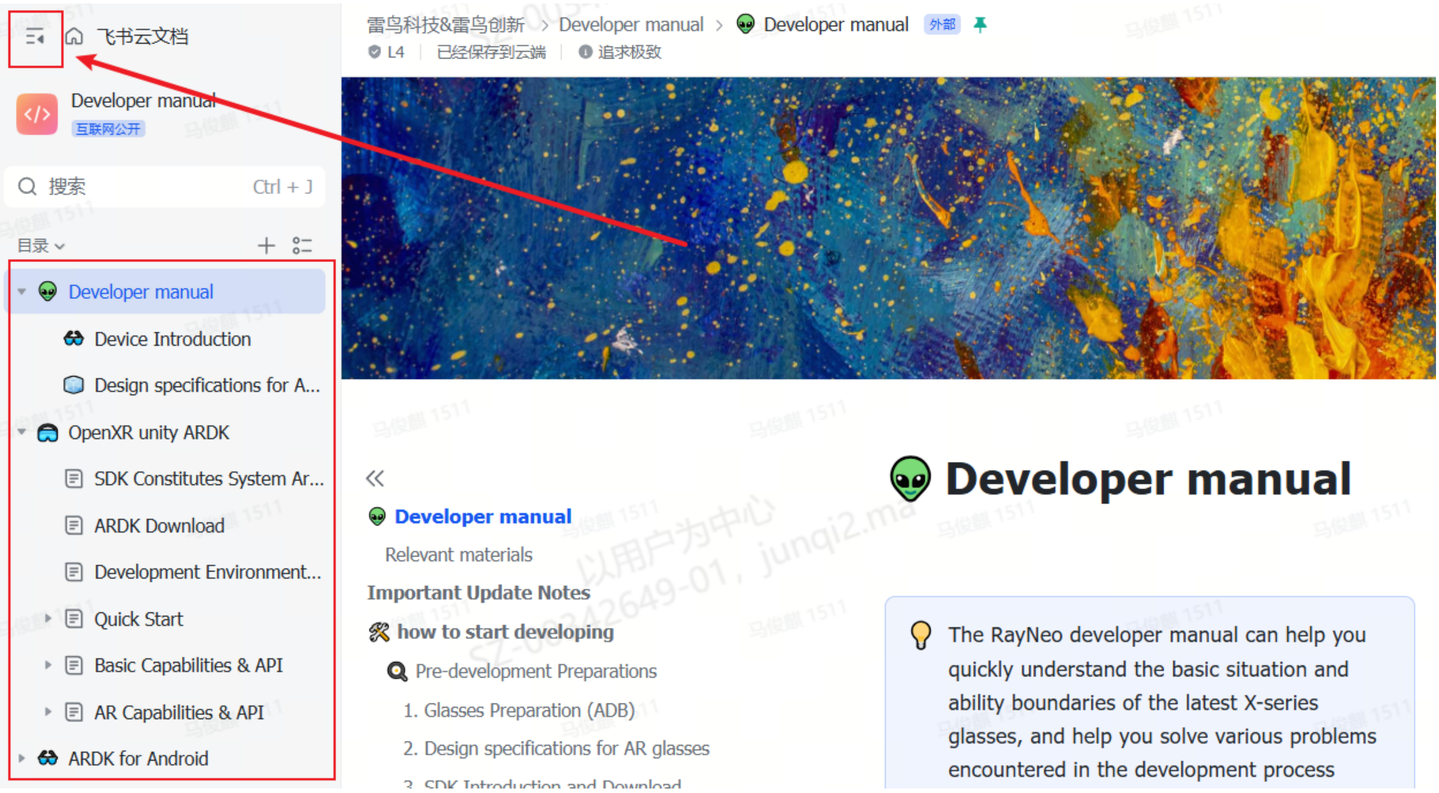
🛠️ how to start developing

🍳 Pre-development Preparations

1. Glasses Preparation (ADB)

> 💡 Since the system and interaction of AR glasses differ significantly from mobile phones, while simulated development is possible, the optimal effect is achieved through direct debugging on the glasses, which can resolve many display and compatibility issues.

- **Glasses**: Prepare an X-series glasses upgraded to the latest OS.
- **Development Manual:** Read to understand the SDK functions (expanded as shown in the figure for easier reading)



- **ADB Connection**: Ensure the glasses are successfully connected to the computer via ADB.
  - The smart glasses support Android ADB commands. For instructions on enabling the required permissions, please refer to: ADB
- **Glasses Screen Mirroring to Computer**: You can download AnLink (for Windows only) or use Scrcpy for screen - mirroring.

2. Design specifications for AR glasses

> 💡 Due to the display mode (OST) and control interaction of AR glasses, there are significant differences with mobile phones, so there are specific specifications on interaction and UI specifications. In order for users to get better and more standardized visual effects and interaction experience on glasses, it is recommended to Read the **design specifications** before development.

- Provides a comprehensive and standard introduction to Interaction Design and Visual Identity specifications.
- Design specifications for fov display area, 3dof freedom, display distance, etc. can help developers understand the default interaction patterns and basic definitions of the system, and develop software based on these interactions.
- The Visual Identity specification can help developers achieve better and more standardized visual effects on AR glasses, ensuring efficient, intuitive, and consistent User Experience.

Details please refer to: Design specifications for AR glasses

3. SDK Introduction and Download

> 💡 Because AR glasses need **dual-screen closing, touchpad control, glasses sensor data acquisition, AR capability provision,** etc., it is necessary to access RayNeoARDK to better present application functions

The official currently only supports two SDKs, Android and Unity (i.e. RayNeo ARDK). Please choose according to the application function.

1. **OpenXR Unity ARDK** **OpenXR unity ARDK**

   Unity provides AR capabilities such as 3DOF, SLAM, and plane detection, making it more suitable for application scenarios with spatial effects.

- Samples in the documentation include example codes for most capabilities; if a capability is not mentioned, it cannot be provided via ARDK for the time being.
- For data like audio, camera, and gyroscope, Android native APIs can be used directly.
- All-in-one glasses have high power consumption requirements; please control the app's running power consumption as much as possible.
- Partial low-power capabilities of AR tools like Vuforia can be used on X2, but overall usage is limited by temperature and power consumption; not recommended.

2. **ARDK for Android** 📄 ARDK for Android

   Android mainly provides basic capabilities such as screen merging, touch control, GPS data, and ring quaternion, suitable for developing information prompt (HUD) applications.

   - The Android SDK is divided into two parts: ARDK for resolving screen merging and control functions, and IPCSDK for providing GPS data, ring customization, etc.
   - Samples in the documentation include example codes for most capabilities; if a capability is not mentioned, it cannot be provided via ARDK for the time being.
   - For data like audio, camera, and gyroscope, Android native APIs can be used directly.

🍷 Suitable Application Scenarios for X-series Glasses

> 💡 Combining the characteristics of X-series all-in-one design, high light transmittance, portability, wide-angle camera, and low power consumption:

Consumer-side scenarios

1. **Tools**: Recommended for utility applications in lifestyle services or work efficiency, such as teleprompters, readers, to-do managers, timers, eye care tools, photo albums, etc.
2. **AI+AR**: Application tools that bring new experiences based on AI large model capabilities and AR features, such as AI assistants, oral language training, blind assistance, communication assistants, etc.
3. **Application Transformation**: Migrate suitable mobile applications to AR glasses (e.g., AI assistants, Mi Home), or applications that can cover user scenarios more comprehensively (e.g., Dida Checklist, flomo).
4. **Others**: Your creativity matters...

Business-side scenarios

Background: Current 6DOF capabilities have high power consumption and are not recommended for long-term use. In B-end scenarios with high availability requirements, it is recommended to use 3DOF or 0DOF scenarios, triggering content through image recognition, GPS positioning, voice, touch, etc., and presenting it via text, voice, and 3D models.

1. **Currently suitable trial scenarios**:
    - Museum and scenic spot navigation
    - Industrial scenario tutorials, step-by-step guidance, remote collaboration, etc.
    - Facial recognition and labeling
    - Patrol record check-in, behavior norms, etc.
    - Others (numerous B-end scenarios, not limited)

2. **Introduction to Capabilities Suitable for Long-Term Use on Glasses**

- **General Interfaces**: Microphone, speaker, camera
- **Provided by SDK**: Eye-closed rendering, 3DoF (3 Degrees of Freedom), GPS data, face detection, gaze ray, image recognition
- **Third-Party Capabilities Required for Integration**: ASR , TTS, LLM, general intelligent capabilities (e.g. image recognition, OCR, etc.)

🪐 How to publish an app

> 💡 If you want to open your app to all RayNeo glasses users, you can submit your app through RayNeo Open Platform. After approval, users will be able to download and install your app in the mobile app that comes with the glasses

1. **Open Platform** : open.rayneo.com
2. **Real-name authentication** : individuals (ID card) or enterprises (business license) can choose one authentication, generally will be reviewed within 5-10 working days, after submission can continue to create applications, without waiting for review

1. **Data submission** : The data introduction will be displayed on the supporting app (RayNeoAR), and the content of other applications can be referred to

2. **Shelf test** : In order to provide a better user experience, we will conduct a complete test in terms of function, interaction, ui, power consumption, etc. If there are any modification suggestions, we will communicate with you
3. **Progress inquiry:** If you need to know about related issues, you can join our developer discord

🙋 how to get development support

1. Official Developer Community

If you have any development-related questions, you can join the communication.

https://discord.gg/7CASXNQG7k

Email： developer@rayneo.com

2. Other

store email： service@rayneo.com

User discord: https://discord.gg/KdBGtpNHux